

The Pol-InSAR Course

Part 11: TomoSAR 2

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

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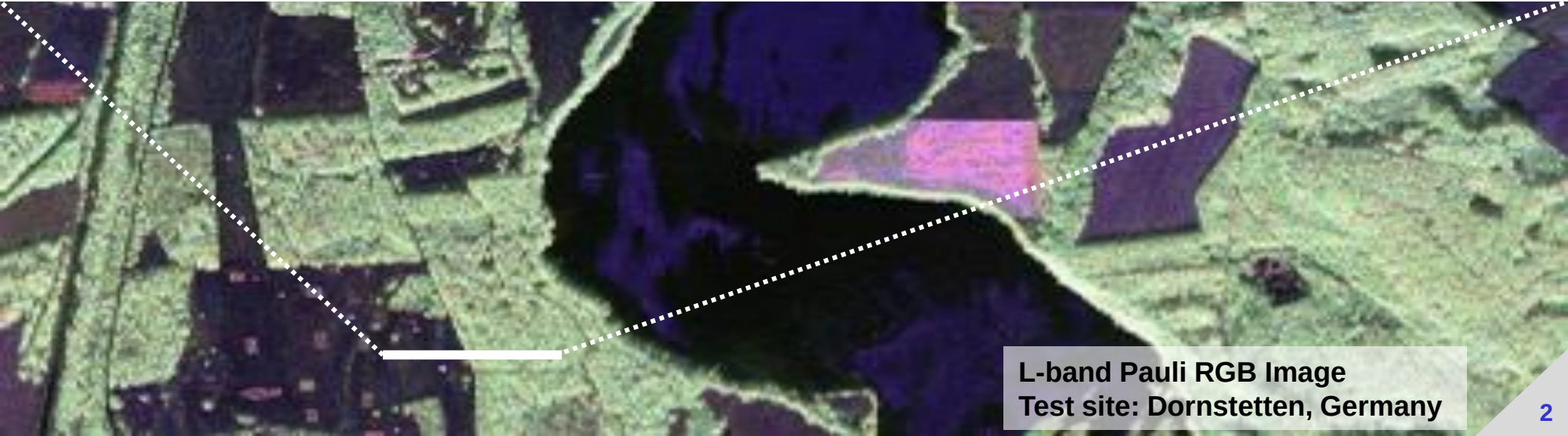
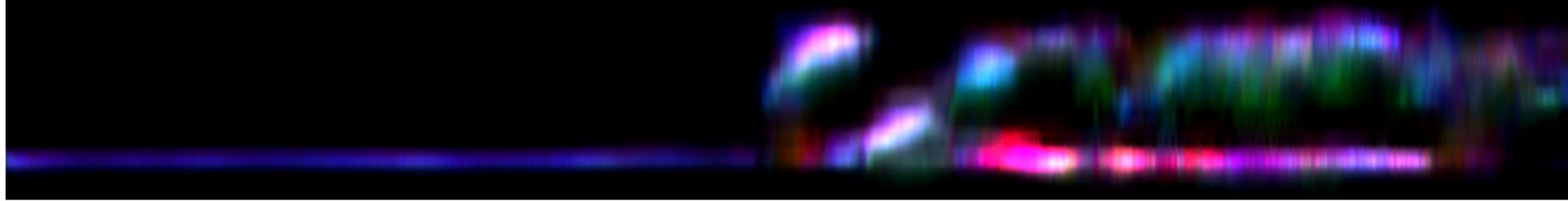


SAR Tomography (TomoSAR)

Vertical Reflectivity Profile (HH)

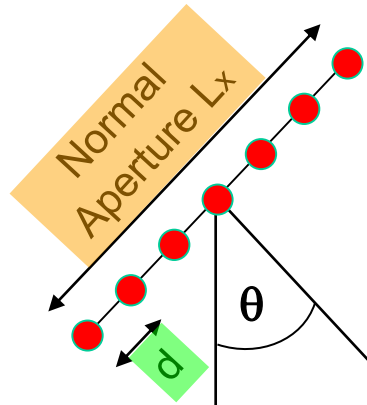


Vertical Reflectivity Profile (Pauli)



L-band Pauli RGB Image
Test site: Dornstetten, Germany

To summarize

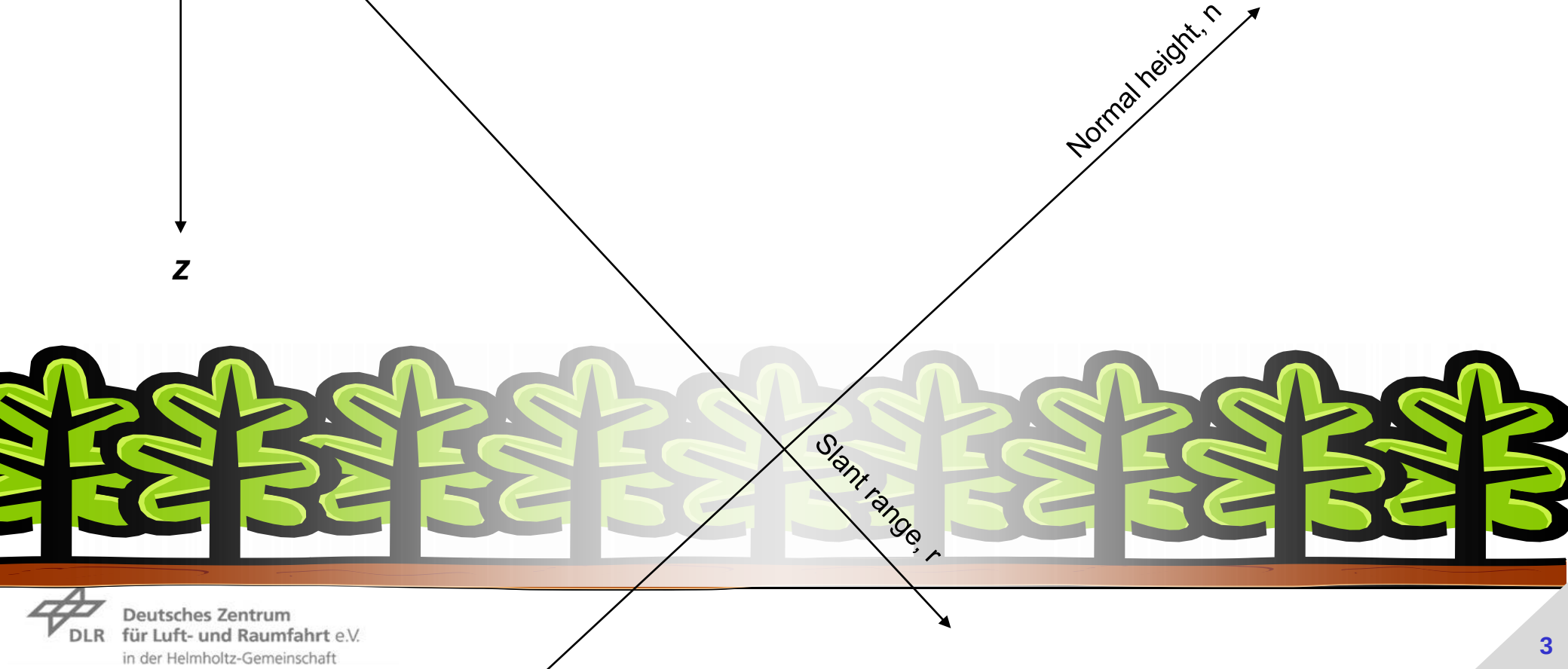


Sampling condition 4 **ambiguity-free** reconstruction within H_V : $d \leq \frac{\lambda}{2H_V} R_0$

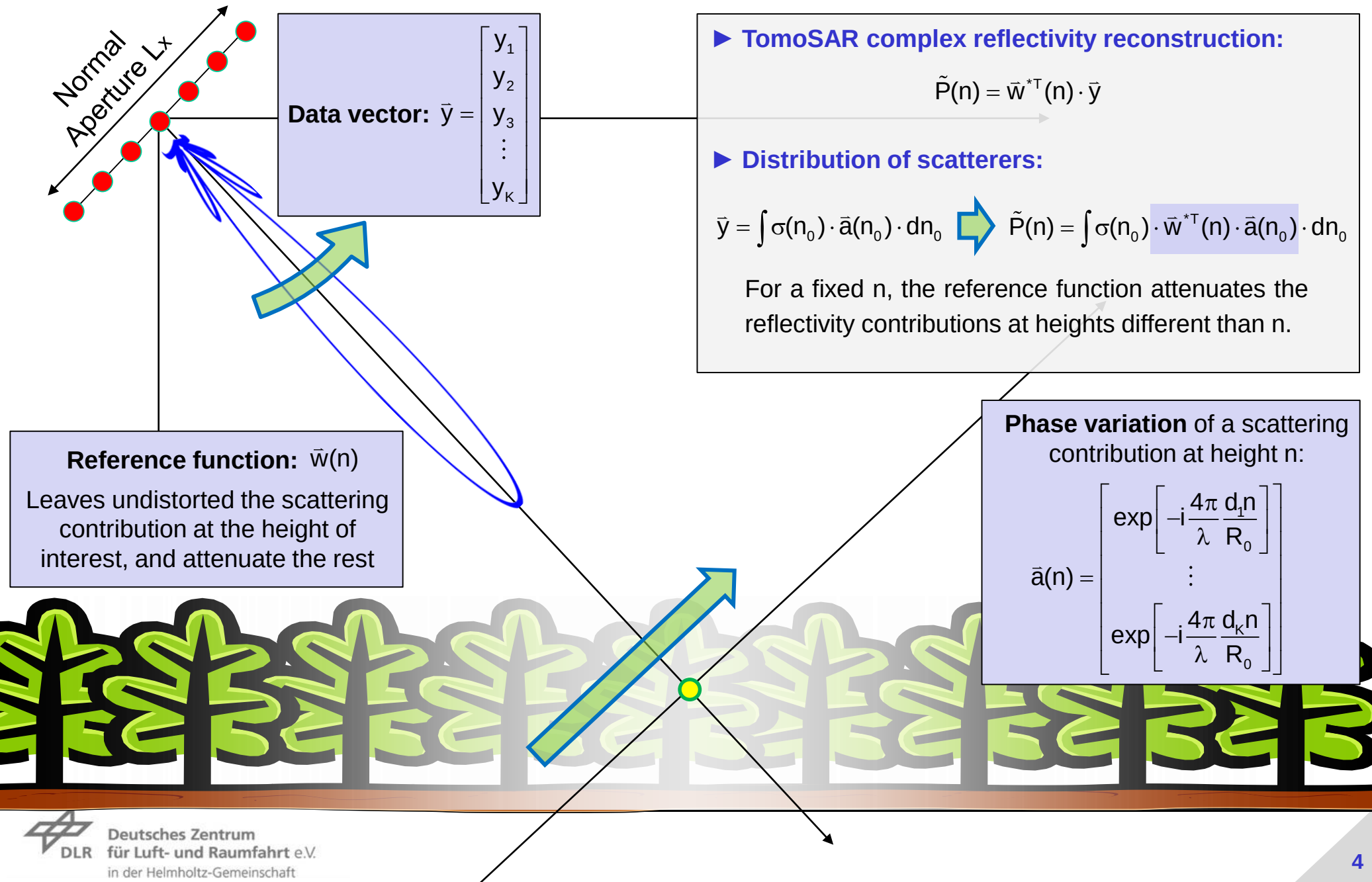
Normal aperture required for a **height resolution** δ_n : $L_x \geq \frac{\lambda}{2\delta_n} R_0$

Number of acquisitions with **uniform** displacements :

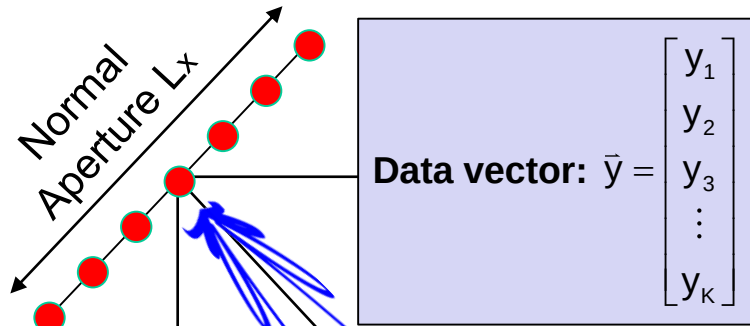
$$K \geq \frac{L_x}{d} + 1$$



Complex reflectivity reconstruction



Complex reflectivity reconstruction



► “Fourier” complex reflectivity reconstruction:

$$\bar{w}(n) = \frac{1}{K} \bar{a}(n) \Rightarrow \tilde{P}(n) = \frac{1}{K} \bar{a}^{\text{T}}(n) \cdot \bar{y}$$

► Distribution of scatterers:

Point-spread function

$$\bar{y} = \int \sigma(n_0) \cdot \bar{a}(n_0) \cdot dn_0 \Rightarrow \tilde{P}(n) = \int \sigma(n_0) \cdot \text{PSF}(n - n_0) \cdot dn_0$$

Only the TomoSAR displacements determine the filtering capabilities of the reference functions.

Reference function: $\bar{w}(n)$

Leaves undistorted the scattering contribution at the height of interest, and attenuate the rest

Phase variation of a scattering contribution at height n :

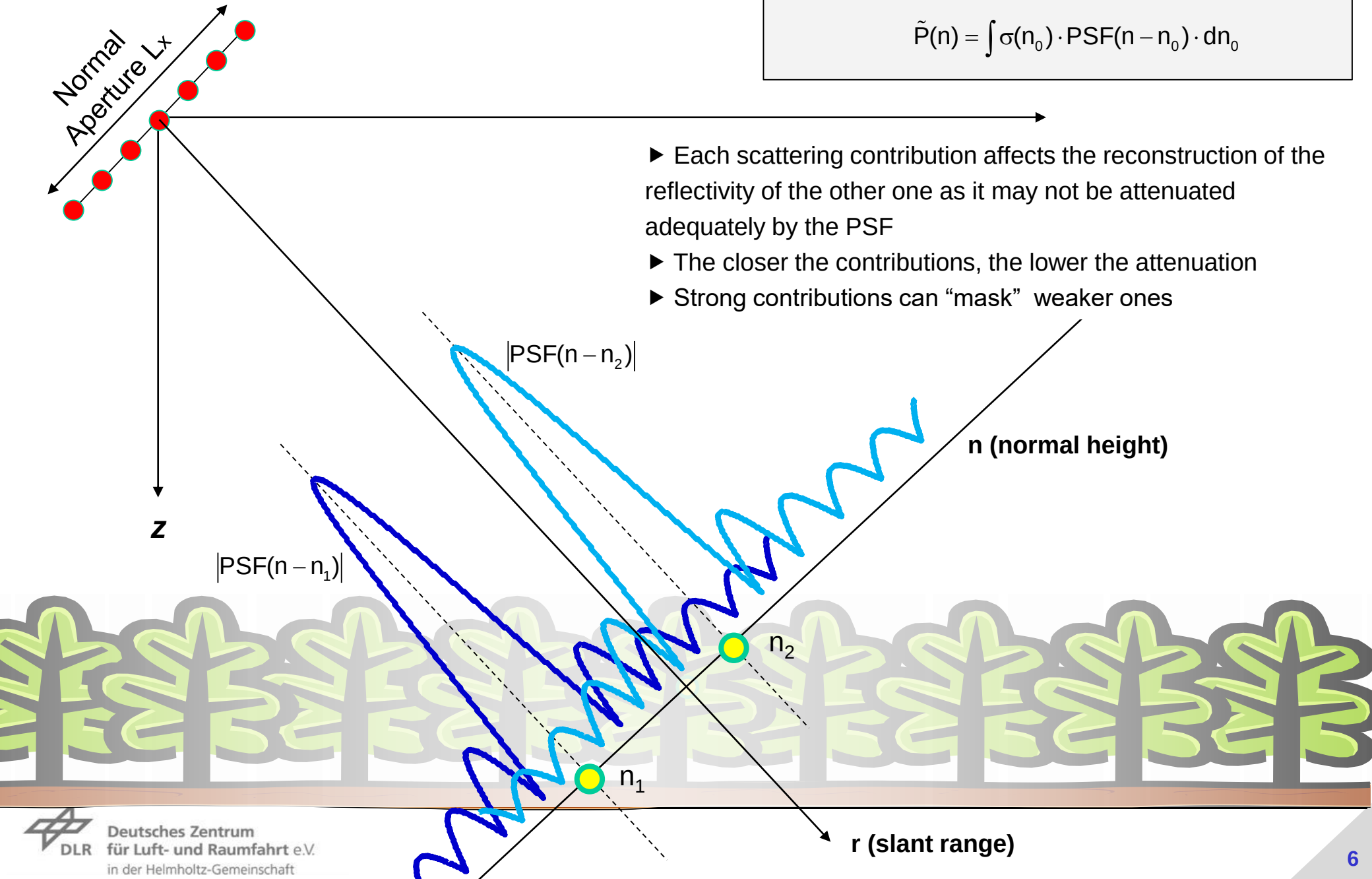
$$\bar{a}(n) = \begin{bmatrix} \exp\left[-i \frac{4\pi}{\lambda} \frac{d_1 n}{R_0}\right] \\ \vdots \\ \exp\left[-i \frac{4\pi}{\lambda} \frac{d_K n}{R_0}\right] \end{bmatrix}$$

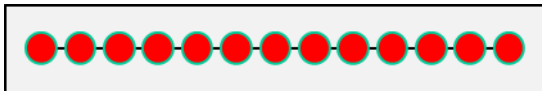
Complex reflectivity reconstruction

► “Fourier” complex reflectivity reconstruction:

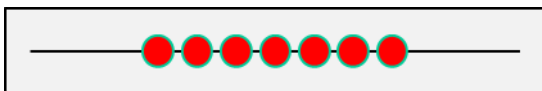
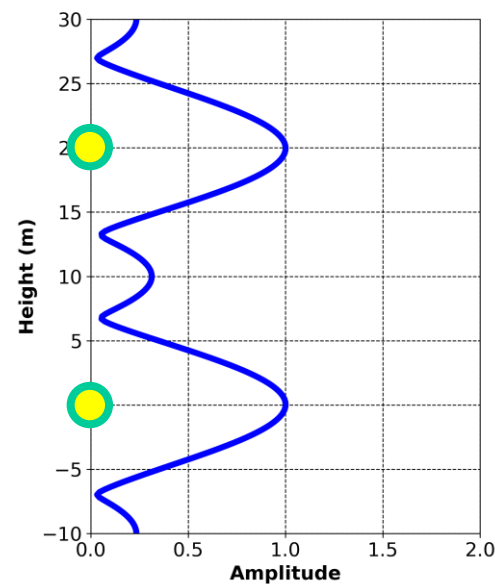
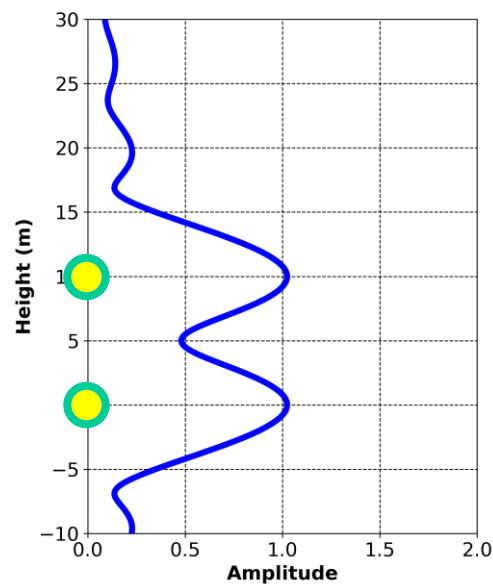
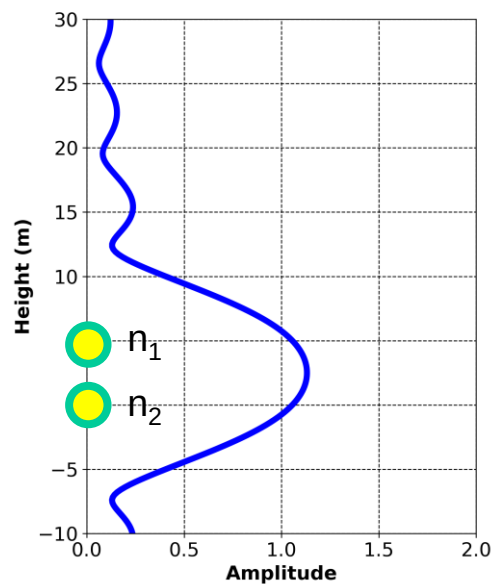
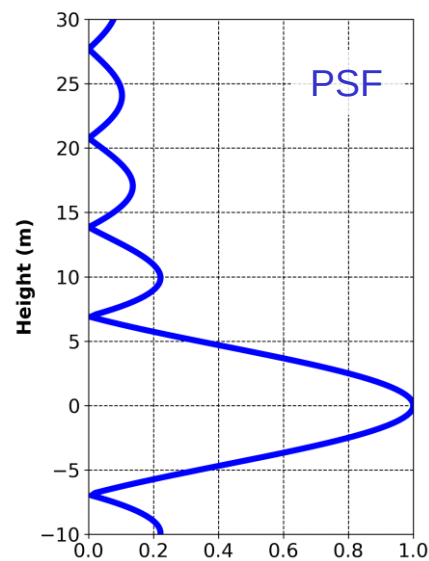
$$\tilde{P}(n) = \int \sigma(n_0) \cdot \text{PSF}(n - n_0) \cdot dn_0$$

- Each scattering contribution affects the reconstruction of the reflectivity of the other one as it may not be attenuated adequately by the PSF
- The closer the contributions, the lower the attenuation
- Strong contributions can “mask” weaker ones

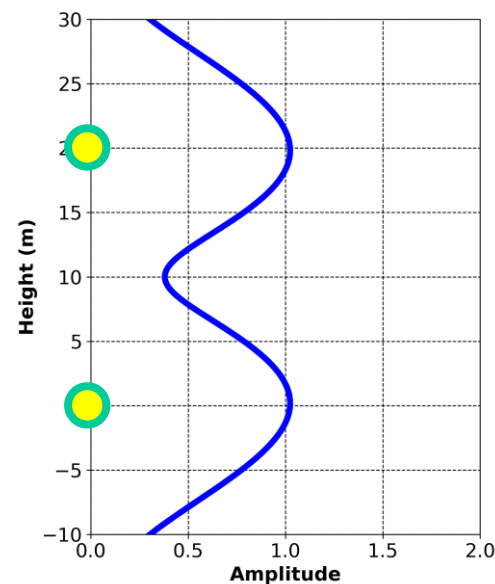
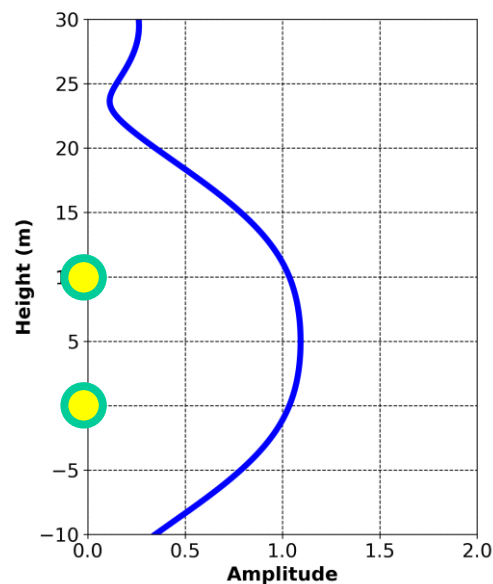
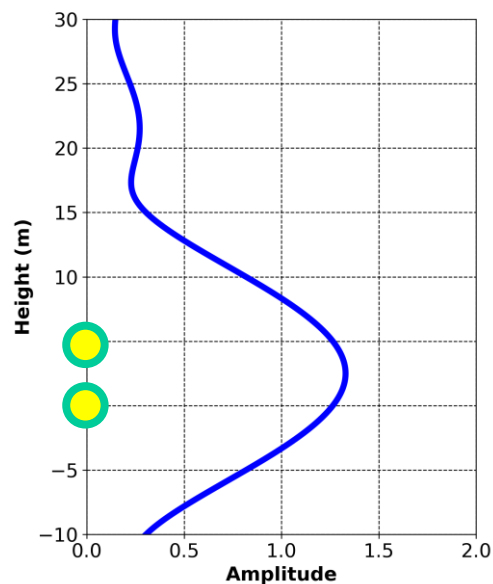
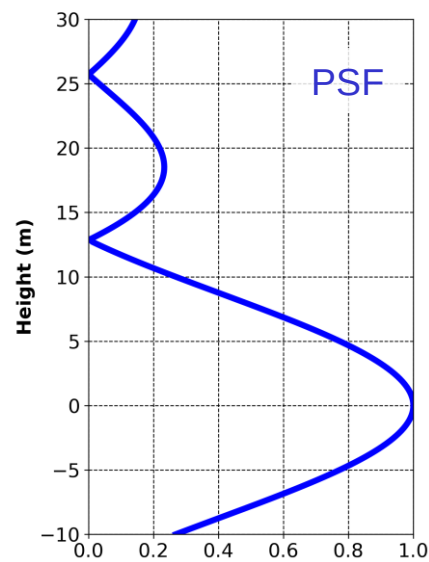




Ambig. free height interval 80 m, 13 unif. acquisitions ► Height resolution 6.67 m



Ambig. free height interval 80 m, 7 unif. acquisitions ► Height resolution 13.33 m

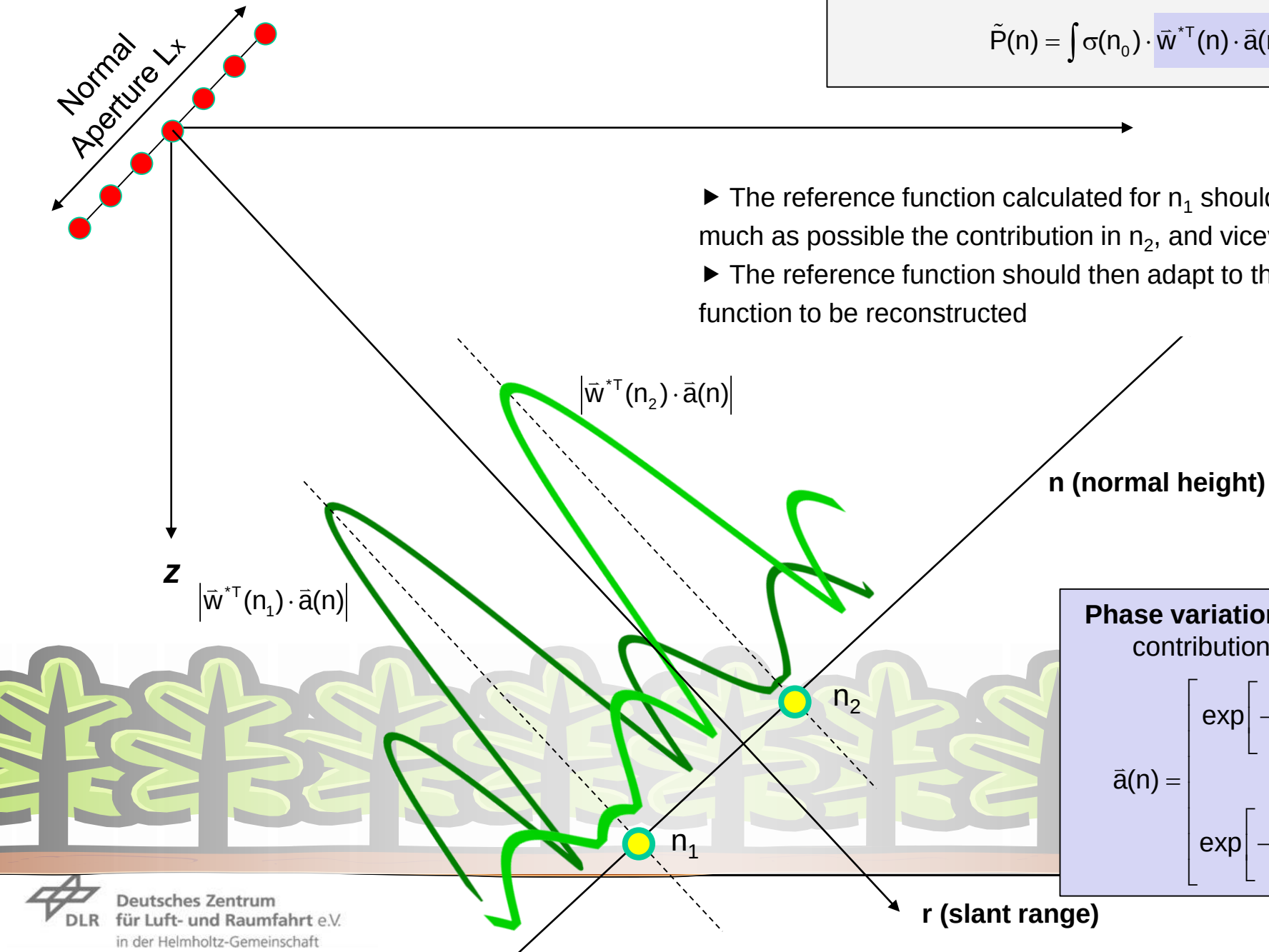


Complex reflectivity reconstruction

► Complex reflectivity reconstruction:

$$\tilde{P}(n) = \int \sigma(n_0) \cdot \bar{w}^{*T}(n) \cdot \bar{a}(n_0) \cdot dn_0$$

- The reference function calculated for n_1 should attenuate as much as possible the contribution in n_2 , and viceversa
- The reference function should then adapt to the reflectivity function to be reconstructed



Phase variation of a scattering contribution at height n :

$$\bar{a}(n) = \begin{bmatrix} \exp\left[-i \frac{4\pi}{\lambda} \frac{d_1 n}{R_0}\right] \\ \vdots \\ \exp\left[-i \frac{4\pi}{\lambda} \frac{d_k n}{R_0}\right] \end{bmatrix}$$

The covariance matrix of a TomoSAR data vector

$$\begin{aligned}
 y_k &= \int \sigma(n) \cdot \exp \left[-i \frac{4\pi}{\lambda} \frac{d_{\perp k} n}{R_0} \right] dn \\
 y_j &= \int \sigma(n) \cdot \exp \left[-i \frac{4\pi}{\lambda} \frac{d_{\perp j} n}{R_0} \right] dn
 \end{aligned}$$

Generic covariance:

$$[R_y]_{k,j} = E\{y_k y_j^*\} = \int E\{\sigma(n)^2\} \cdot \exp \left[-i \frac{4\pi}{\lambda} \frac{(d_{\perp k} - d_{\perp j}) n}{R_0} \right] dn$$

Interferometric coherence:

$$\gamma_{k,j} = \frac{[R_y]_{k,j}}{\sqrt{[R_y]_{k,k} [R_y]_{j,j}}}$$

Vertical height: $z = n \sin \theta$

$$[R_y]_{k,j} = \int f(z) \cdot \exp[-i (k_{z,k} - k_{z,l}) z] dz$$

Vertical reflectivity function: $f(z) = E\{\sigma(n)^2\} / \sin \theta$

Vertical wavenumber: $k_z \approx \frac{4\pi}{\lambda R_0} \frac{d_{\perp}}{\sin \theta}$

► Covariance matrix:

$$[R_y] = E\{\bar{y} \bar{y}^T\} = \begin{bmatrix} E\{y_1 y_1^*\} & E\{y_1 y_2^*\} & \dots & \dots & E\{y_1 y_K^*\} \\ E\{y_2 y_1^*\} & \ddots & & & \vdots \\ \vdots & & E\{y_k y_j^*\} & & \vdots \\ \vdots & & & \ddots & E\{y_{K-1} y_K^*\} \\ E\{y_K y_1^*\} & \dots & \dots & E\{y_K y_{K-1}^*\} & E\{y_K y_K^*\} \end{bmatrix}$$

► Reconstruction of $f(z)$:

$$E\left\{\left|\tilde{P}(n)\right|^2\right\} = \bar{w}^*{}^T(n) \cdot E\{\bar{y} \bar{y}^T\} \cdot \bar{w}(n) = \bar{w}^*{}^T(n) \cdot [R_y] \cdot \bar{w}(n)$$

TomoSAR intensity profile

Vertical reflectivity reconstruction: Capon algorithm

- **Design criterion:** Attenuate as much as possible the reflectivity of the scattering components at heights different from n , while leaving undistorted the component at n :

$$\min \bar{\mathbf{w}}^{*T}(n) \cdot [\mathbf{R}_y] \cdot \bar{\mathbf{w}}(n) \quad \text{subject to} \quad \bar{\mathbf{w}}^{*T}(n) \cdot \bar{\mathbf{a}}(n) = 1$$

$$\bar{\mathbf{w}}(n) = \frac{[\mathbf{R}_y]^{-1} \cdot \bar{\mathbf{a}}(n)}{\bar{\mathbf{a}}^{*T}(n) \cdot [\mathbf{R}_y]^{-1} \cdot \bar{\mathbf{a}}(n)}$$

The reference function adapts to (1) the underlying vertical reflectivity profile, and (2) its individual reflectivity components.

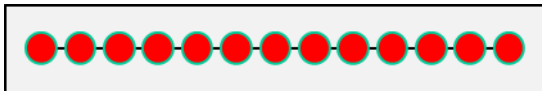
- If $[\mathbf{R}_y] = [\mathbf{I}]$ (identity matrix) is assumed in the filter design:

$$\min \bar{\mathbf{w}}^{*T}(n) \cdot \bar{\mathbf{w}}(n) \quad \text{subject to} \quad \bar{\mathbf{w}}^{*T}(n) \cdot \bar{\mathbf{a}}(n) = 1$$

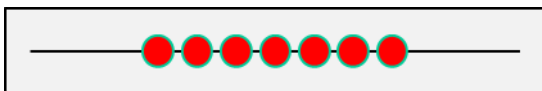
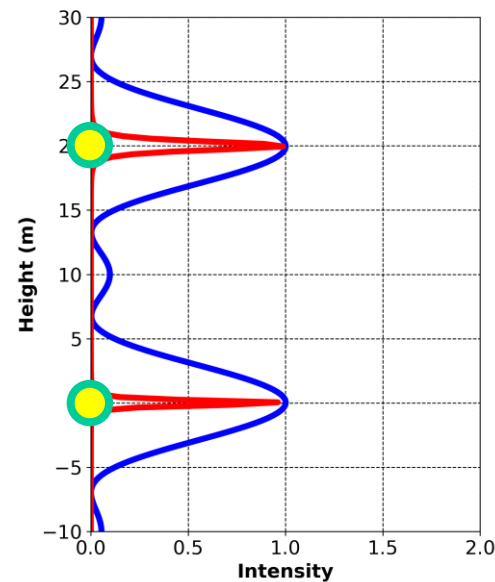
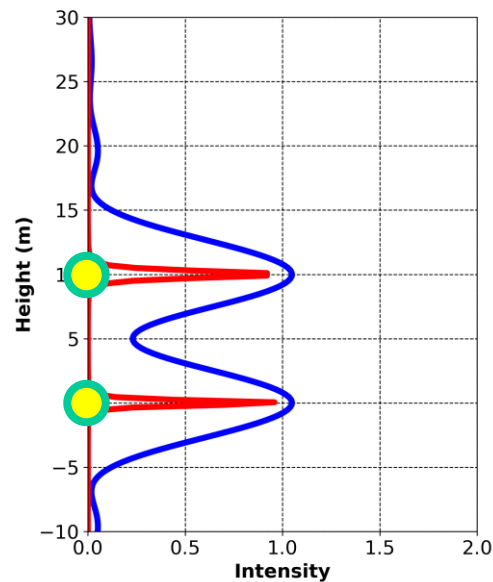
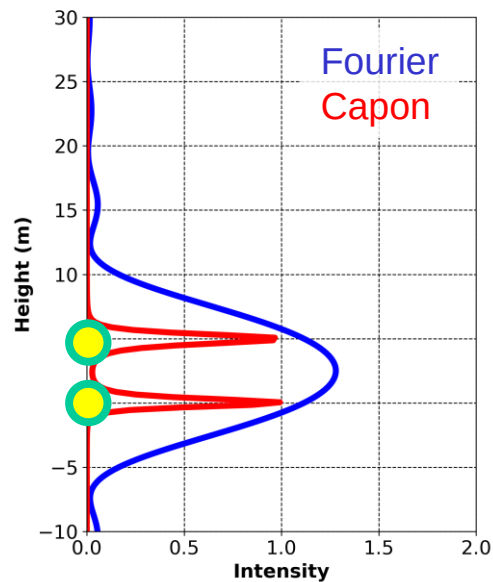
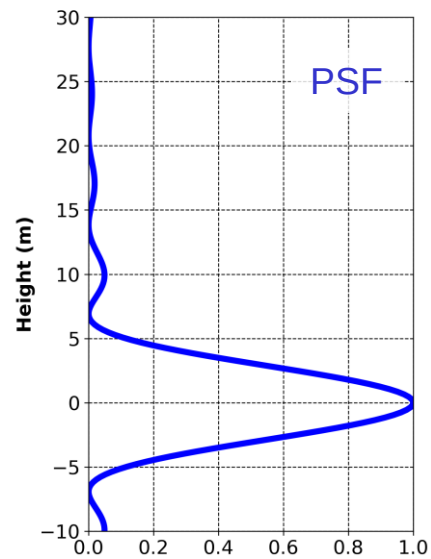
$$\bar{\mathbf{w}}(n) = \frac{1}{K} \bar{\mathbf{a}}(n)$$

The Fourier reference function treats all the height in same way, independently of the effective presence of a scattering component.

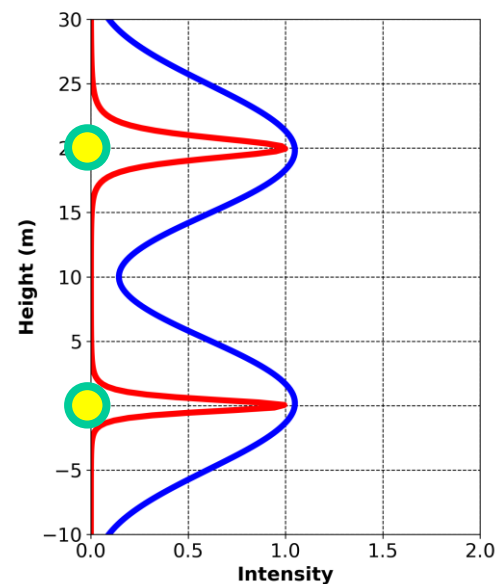
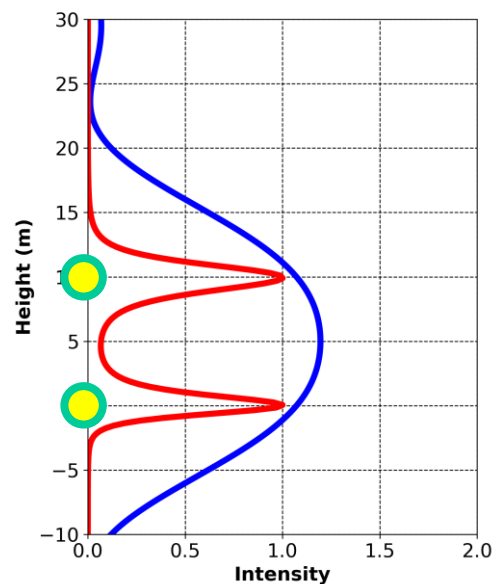
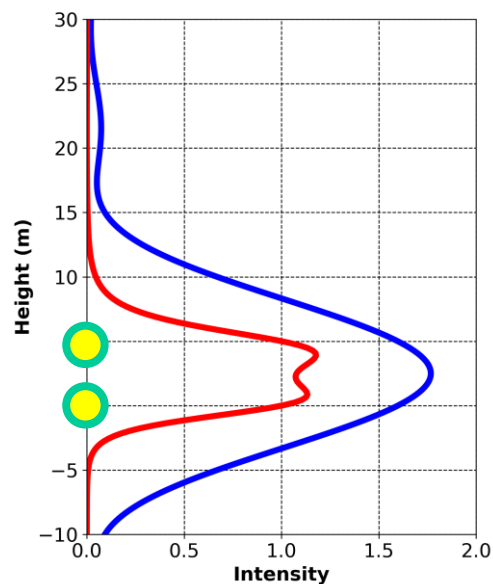
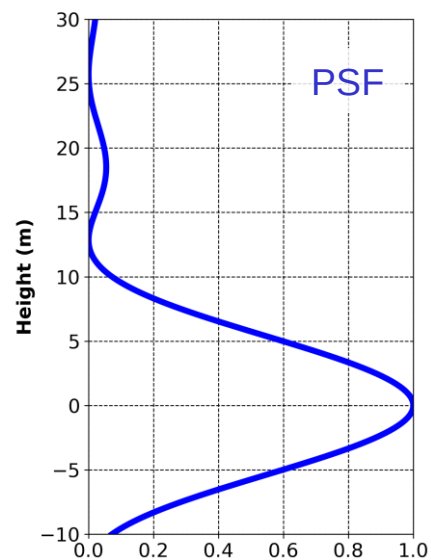




Ambig. free height interval 80 m, 13 unif. acquisitions ► Height resolution 6.67 m

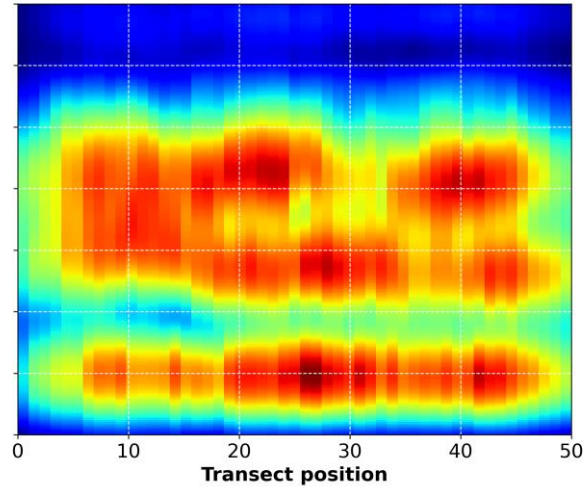
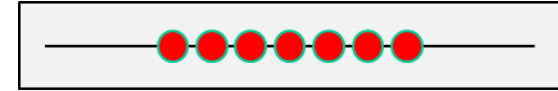
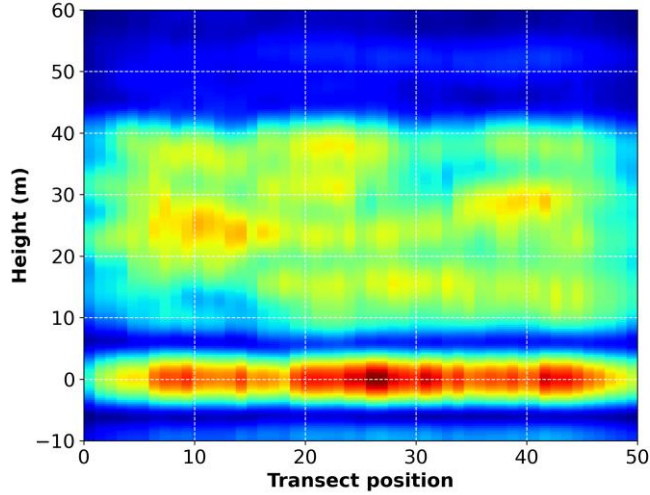
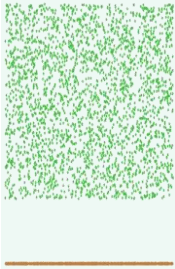


Ambig. free height interval 80 m, 7 unif. acquisitions ► Height resolution 13.33 m

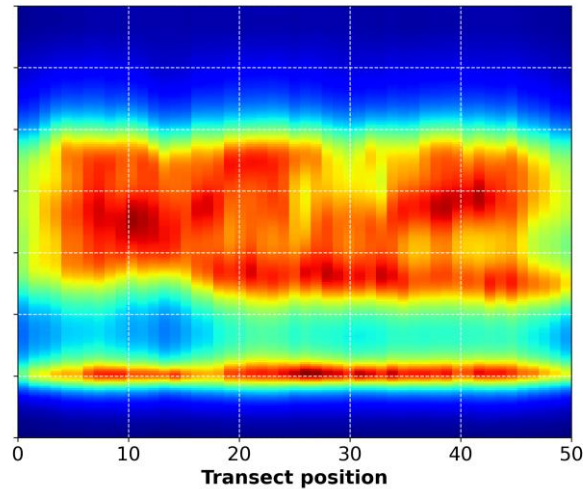
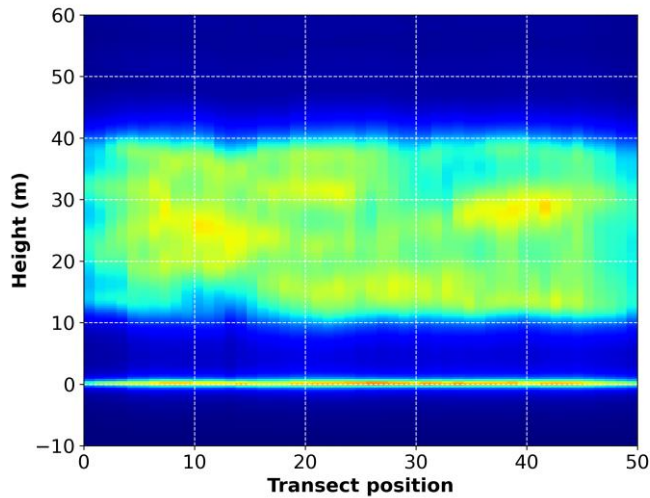
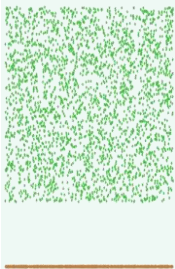


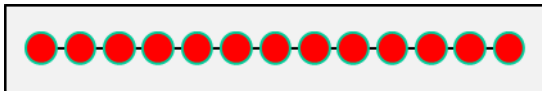
Vertical reflectivity reconstruction of volumes

Fourier

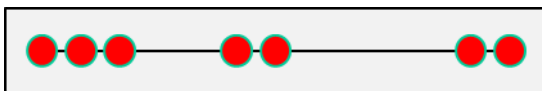
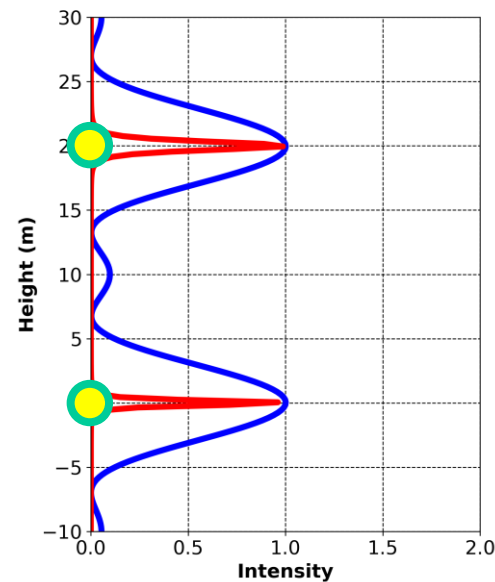
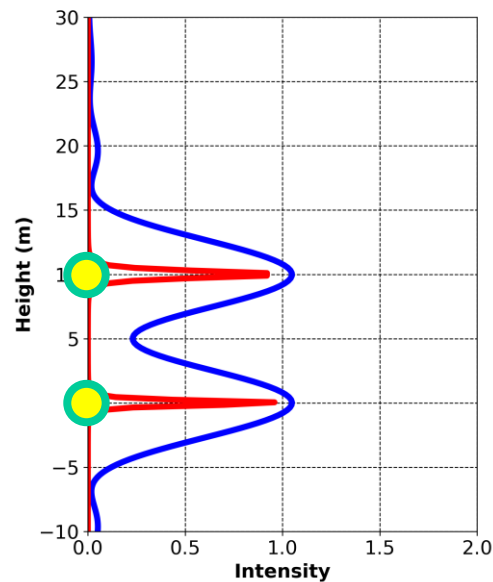
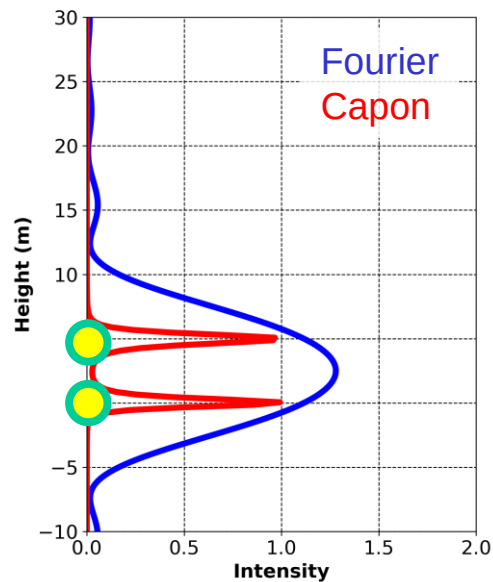
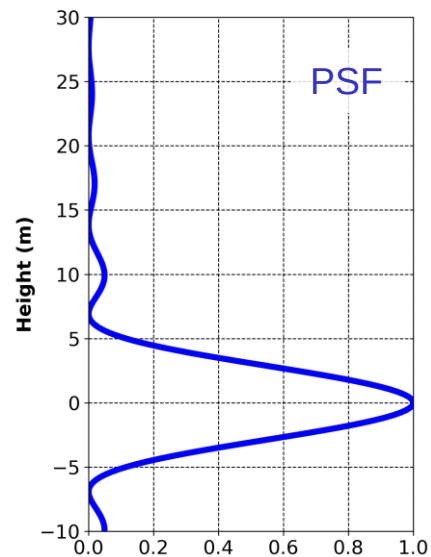


Capon

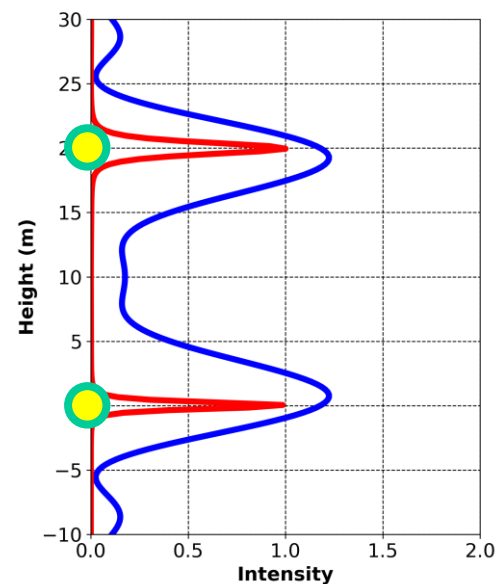
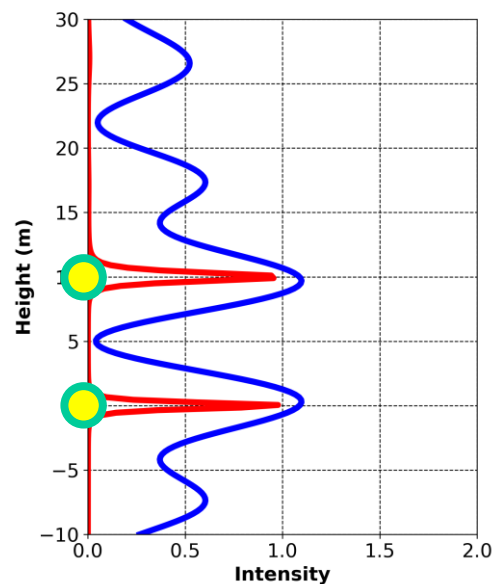
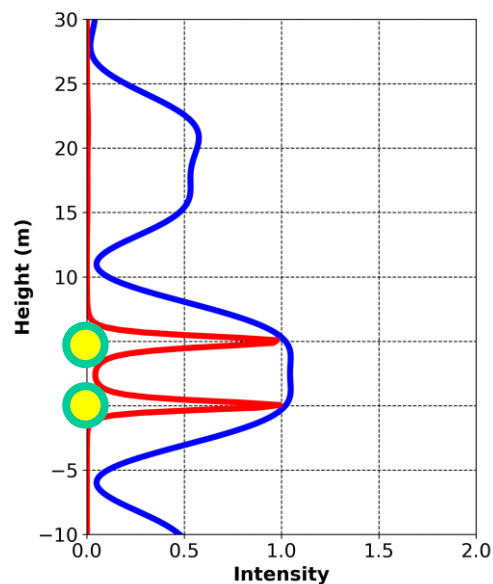
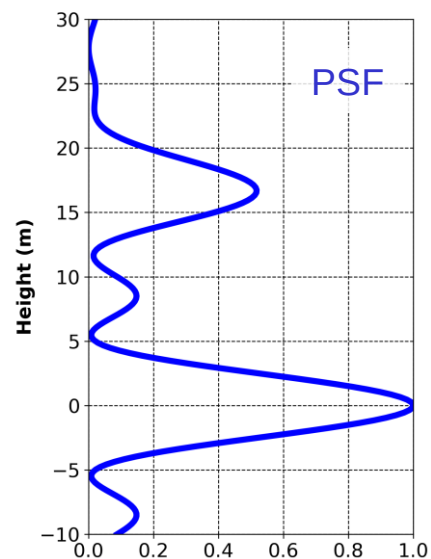




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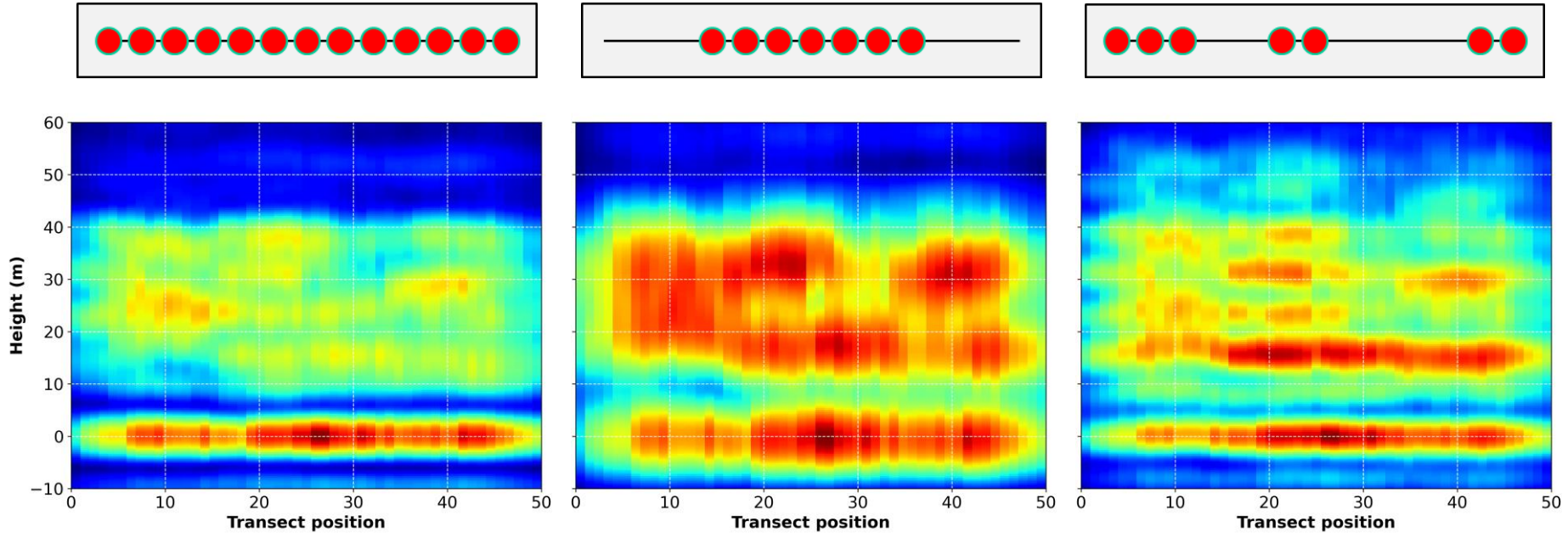
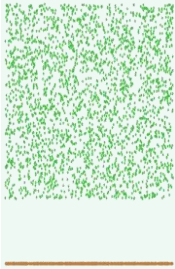


Ambig. free height interval 80 m, height resolution 13.33 m, 7 non-unif. acquisitions

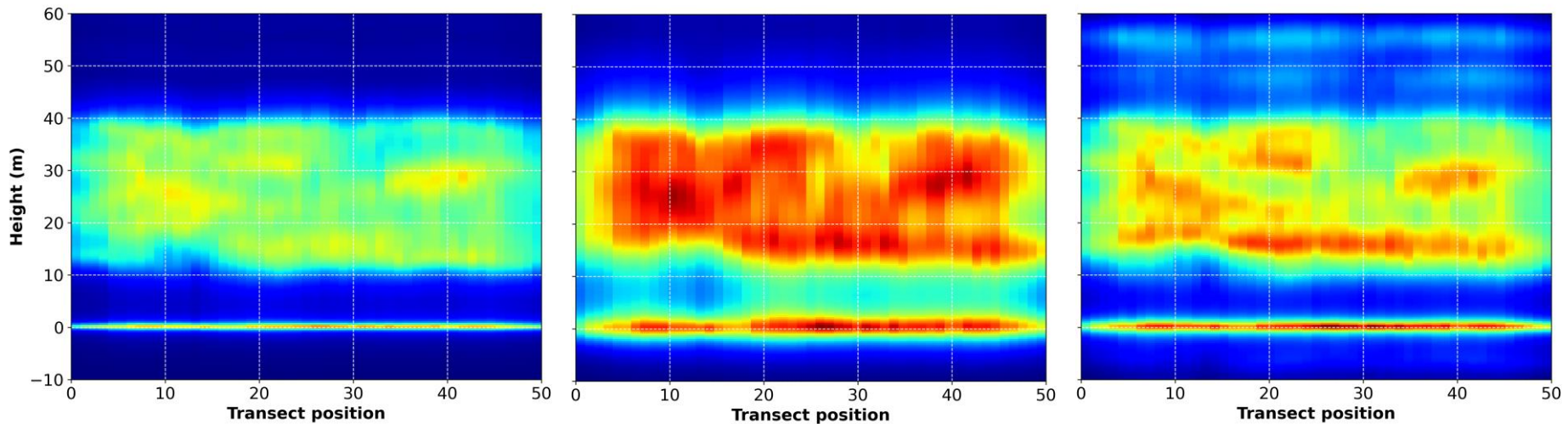
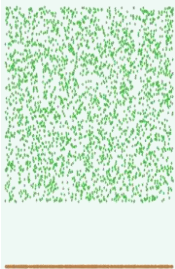


Vertical reflectivity reconstruction of volumes

Fourier



Capon



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